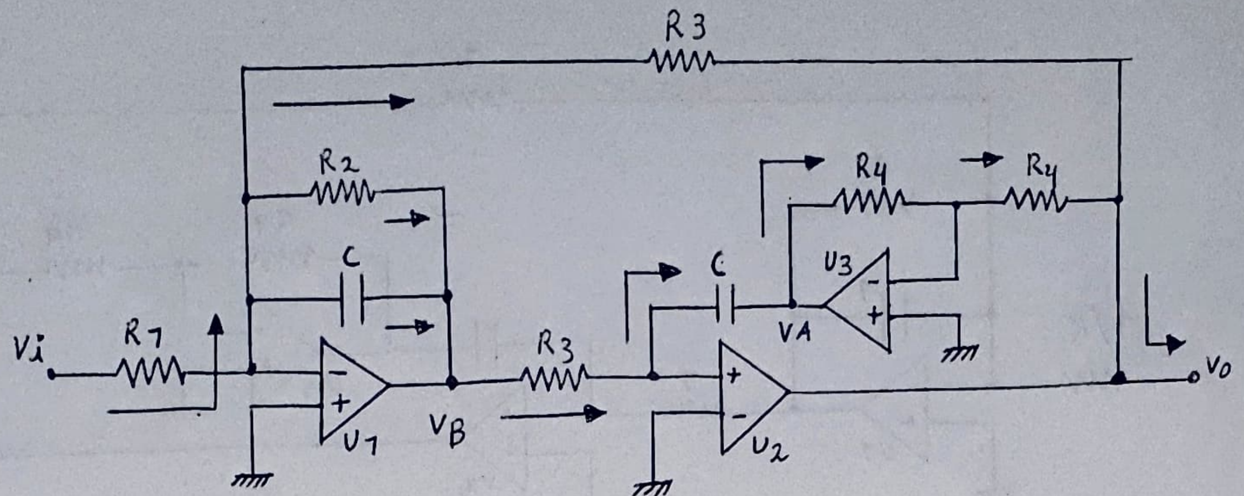
TS2



$$\boxed{H(S) = \frac{V_O}{V_i}} \quad (1) \qquad ; \qquad V_A \cdot \cancel{G_4} = -V_O \cdot \cancel{G_4} \longrightarrow \boxed{V_A = -V_O} \quad (2)$$

$$\boxed{V_B \cdot G_3 = -SC \cdot V_A} \quad (3) \qquad ; \qquad \boxed{V_i \cdot G_1 = -V_B \cdot SC - V_B G_2 - V_O G_3} \quad (4)$$

$$(2) \rightarrow (3): \quad V_B \cdot G_3 = \cancel{-}SC \cdot (\cancel{-}V_O) \longrightarrow \boxed{V_B = V_O \cdot \frac{SC}{G_3}} \quad (5)$$

$$(5) \rightarrow (4): \quad V_i \cdot G_1 = -V_O \cdot \frac{SC}{G_3} \cdot SC - V_O \frac{SC}{G_3} \cdot G_2 - V_O \cdot G_3$$

$$V_i \cdot G_1 = -V_O \cdot \left(\frac{s^2 c^2}{G_3} + \frac{SC \cdot G_2}{G_3} + G_3 \right)$$

$$V_i \cdot G_1 = -V_O \cdot \left(\frac{s^2 c^2 + SC \cdot G_2 + G_3^2}{G_3} \right)$$

$$\longrightarrow H(S) = \frac{V_O}{V_i} = -\frac{G_1}{G_3} \cdot \frac{G_3^2}{c^2} \cdot \frac{1}{s^2 + s\frac{G_2}{C} + \frac{G_3^2}{C^2}}$$

$$\boxed{H(S) = -\frac{G_1}{G_3} \cdot \frac{\frac{G_3^2}{C^2}}{s^2 + s\frac{G_2}{C} + \frac{G_3^2}{C^2}}}$$